This section completed the modelling of the projection with central cameras. Estimating all parameters of the interior orientation will be discussed in Sects. 15.4 and 15.5, p. 696 on self-calibrating bundle adjustment and calibration, respectively, where we also discuss how to decide on the adequate model to be used.

12.2.4 Spatial Resection

If the interior orientation of the camera is known and thus calibrated, we may determine the six parameters of the exterior orientation of a single image by a *spatial resection* from a set of observed scene points or lines.

spatial resection, PnP problem

In the computer vision community this sometimes is called the *perspective n-point problem* (PnP problem). It assumes n corresponding points are given in 2D and 3D for determining the relative pose of the camera and scene. This suggests that also the pose of the scene w.r.t. the camera could be determined using the spatial resection. An example for both tasks is given in Fig. 12.19.

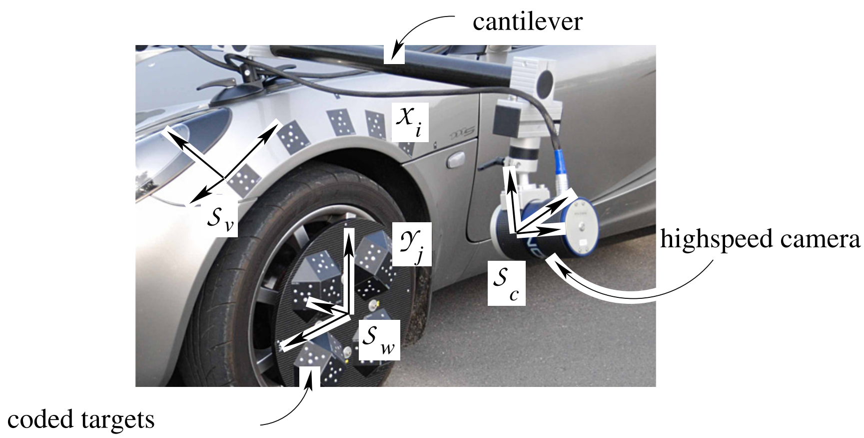

Fig. 12.19 The principle of the system *Wheelwatch* of Aicon for monitoring the rotational motion of a wheel at high speeds. The relative pose of the wheel coordinate system S_w w.r.t. the vehicle system S_v via the camera system S_c with a high speed camera rigidly fixed to the car using a cantilever is based on the image coordinates in each image showing both targeted points on the car body and targeted points on the rotating wheel. The relative pose of the camera system S_c w.r.t. the vehicle system S_v is first determined by a spatial resection with the point correspondences $(\mathcal{X}_i, \chi_i')$. The relative pose of the wheel coordinate system S_w w.r.t. the camera coordinate system S_c can be determined by a second spatial resection with the correspondences $(\mathcal{Y}_i, y_i')$. The concatenation of the two motions yields the required relative pose of the wheel w.r.t. the car. The establishment of the correspondences is simplified by using coded targets

We discuss a direct solution for observed scene points for the minimal and the redundant case and a statistical solution for observed scene points and lines (cf. Sect. 12.2.4.3). There exists a direct solution of the pose of a calibrated camera from three observed 3D lines (cf. Dhome et al., 1989).

12.2.4.1 Minimum Solution for Spatial Resection from Points

We start with a direct solution with the minimum number of three scene points, which are observed in an image. All solutions start with determining the distances from the projection centre to the three given points. We present the one by Grunert (1841). The second step consists in determining a spatial motion from three points, which is relatively easy.

We assume that three object points $\mathcal{X}_i, i = 1, 2, 3$, are observed. Either we directly obtain the ray directions ${}^c\mathbf{x}'_i$, or, assuming sensor coordinates $\mathbf{x}'_i$ are given, we determine the normalized ray directions ${}^c\mathbf{x}'^s_i := -\text{sign}(c)\mathsf{N}(\mathsf{K}^{-1}\mathbf{x}'_i)$, cf. (12.108), p. 492. With the coordinates $\boldsymbol{Z}$ of the projection centre O, the rotation matrix R, depending on three parameters, and the three distances $d_i = |\boldsymbol{Z} - \boldsymbol{X}_i|$ to the control points, we obtain the relations

$$d_i{}^c\mathbf{x}'^s_i = R(\boldsymbol{X}_i - \boldsymbol{Z}), \qquad i = 1, 2, 3. \tag{12.199}$$

These are nine equations for the nine unknown parameters, the three coordinates of $O(\boldsymbol{Z})$, the three distances d_i, and the three parameters for R.

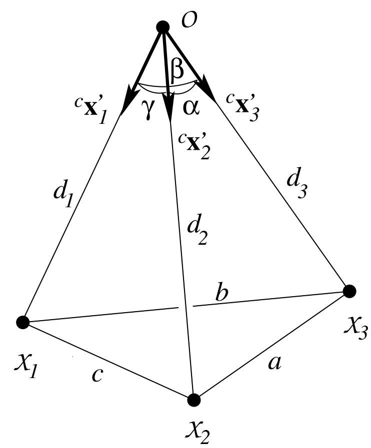

Fig. 12.20 Spatial resection with three given object points, $\mathcal{X}_1$, $\mathcal{X}_2$ and $\mathcal{X}_3$

Step 1: Determination of the Distances. From the 3D points $\mathcal{X}_1$, $\mathcal{X}_2$, and $\mathcal{X}_3$ and from the direction vectors ${}^c\mathbf{x}'_i$ we can first determine the three sides a, b, and c and the three angles α, β, and γ, of the tetrahedron. The law of cosines then yields three constraints for the unknown sides d_i (Fig. 12.20),

$$\begin{aligned} a^2 &= d_2^2 + d_3^2 - 2d_2d_3\cos\alpha \\ b^2 &= d_3^2 + d_1^2 - 2d_3d_1\cos\beta \\ c^2 &= d_1^2 + d_2^2 - 2d_1d_2\cos\gamma\,. \end{aligned} \tag{12.200}$$

These equations are invariant to a common sign of the distances. With the substitutions

$$u = \frac{d_2}{d_1} \qquad v = \frac{d_3}{d_1} \tag{12.201}$$

we obtain from (12.200), solving for d_1^2,

$$d_1^2 = \frac{a^2}{u^2 + v^2 - 2uv\cos\alpha} \tag{12.202}$$

$$= \frac{b^2}{1 + v^2 - 2v\cos\beta} \tag{12.203}$$

$$= \frac{c^2}{1 + u^2 - 2u\cos\gamma}\,. \tag{12.204}$$

These are two constraints in u and v, namely the identity of the terms (12.202) and (12.203) and the identity of the terms (12.203) and (12.204). Therefore we may express u in terms of v quadratically; e.g., from the first two equations and after substitution into the last two equations we obtain a polynomial of degree 4 in v which needs to vanish:

$$A_4 v^4 + A_3 v^3 + A_2 v^2 + A_1 v^1 + A_0 = 0\,. \tag{12.205}$$

The coefficients (Haralick et al., 1994) depend on the known values a, b, c, α, β, and γ:

$$\begin{aligned}
A_4 &= \left(\frac{a^2-c^2}{b^2}-1\right)^2 - \frac{4c^2}{b^2}\cos^2\alpha \\
A_3 &= 4\left\{\frac{a^2-c^2}{b^2}\left(1-\frac{a^2-c^2}{b^2}\right)\cos\beta \right.\\
&\quad \left. + 2\frac{c^2}{b^2}\cos^2\alpha\cos\beta - \left(1-\frac{a^2+c^2}{b^2}\right)\cos\alpha\cos\gamma\right\} \qquad (12.206)\\
A_2 &= 2\left\{\left(\frac{a^2-c^2}{b^2}\right)^2 + 2\left(\frac{a^2-c^2}{b^2}\right)^2\cos^2\beta + 2\left(\frac{b^2-c^2}{b^2}\right)\cos^2\alpha\right.\\
&\quad \left. + 2\left(\frac{b^2-a^2}{b^2}\right)\cos^2\gamma - 4\left(\frac{a^2+c^2}{b^2}\right)\cos\alpha\cos\beta\cos\gamma - 1\right\}\\
A_1 &= 4\left\{-\left(\frac{a^2-c^2}{b^2}\right)\left(1+\frac{a^2-c^2}{b^2}\right)\cos\beta\right.\\
&\quad \left. + \frac{2a^2}{b^2}\cos^2\gamma\cos\beta - \left(1-\left(\frac{a^2+c^2}{b^2}\right)\right)\cos\alpha\cos\gamma\right\} \qquad (12.207)\\
A_0 &= \left(1+\frac{a^2-c^2}{b^2}\right)^2 - \frac{4a^2}{b^2}\cos^2\gamma\,.
\end{aligned}$$

After determination of up to four solutions for v, we obtain the three distances d_i via u: the distance d_1 from (12.203) and (12.204) and d_2 and d_3 from (12.201). *up to four solutions*

Step 2: Exterior Orientation. After having determined the three distances, we need to determine the six parameters of the exterior orientation of the camera.

We first determine the 3D coordinates of the three points in the camera coordinate system,

$${}^c\boldsymbol{X}_i = d_i\,{}^c\mathbf{x}_i'^s\,. \tag{12.208}$$

Then we have to determine the translation $\boldsymbol{Z}$ and the rotation R of the motion between the object and the camera system from

$${}^c\boldsymbol{X}_i = R(\boldsymbol{X}_i - \boldsymbol{Z}) \qquad i = 1, 2, 3\,. \tag{12.209}$$

We can use the direct solution for the similarity transformation from Sect. 10.5.4.3, p. 408, omitting the scale estimate.

Figure 12.21 shows an example with four solutions. A fourth given point correspondence can generally be used to choose the correct solution. If the angles $\angle(OX_1X_2) = \angle(OX_1X_3)$ are close to 90^o, two solutions are very close together. More configurations with multiple solutions are discussed by Fischler and Bolles (1981). A characterization of all multiple solutions is given by Gao et al. (2003). Additional direct solutions for spatial resection are collected in Haralick et al. (1994), cf. also Kneip et al. (2011).

Critical Configuration. As in the case of uncalibrated straight line-preserving cameras, here we also have a critical configuration in which ambiguous or unstable solutions are possible. If the projection centre lies on a circular cylinder, the *dangerous cylinder* (Finsterwalder, 1903), through and perpendicular to the triangle $(\mathcal{X}_1\mathcal{X}_2\mathcal{X}_3)$, the solution is unstable, corresponding to a singular normal equation matrix. In this case two solutions of the fourth-order polynomial are identical. If the projection centre is close to the critical cylinder, the solution is not very precise.

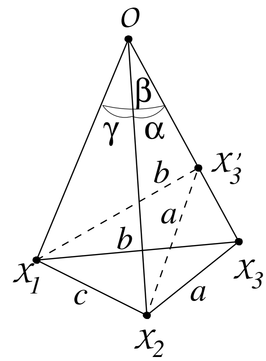

Fig. 12.21 An example of four solutions. For the pyramid shown we have $\alpha = \beta = \gamma$ and $a = b = c$. If we rotate the triangle $\mathcal{X}_1\mathcal{X}_2\mathcal{X}_3$ around $\mathcal{X}_1\mathcal{X}_2$ there exists a position where $\triangle\, \mathcal{X}_1\mathcal{X}_2\mathcal{X}_3 \equiv \triangle\, \mathcal{X}_1\mathcal{X}_2\mathcal{X}_3'$. Thus we have an additional solution. Analogously we find solutions with rotation around $\mathcal{X}_2\mathcal{X}_3$ and $\mathcal{X}_3\mathcal{X}_1$. This is why we have four positions of the triangle with respect to the projection centre and vice versa in this particular configuration

The acceptability of the configuration can again be evaluated by an analysis of the covariance matrix of the unknown parameters with respect to a reference configuration (cf. Sect. 4.6.2.3, p. 120) derived from an iterative solution, as will be shown in the next section.

Direct Determination of the Covariance Matrix of the Projection Centre. After having determined the orientation, we may want to evaluate the solution. This is especially useful if the direct solution is used in a RANSAC procedure and we are interested in evaluating the acceptability of the selected point triple.

The general technique for determining the covariance matrix of a set of unknowns from a set of observations without redundancy has been given in Sect. 2.7.5, p. 43.

Given the nonlinear relation $\boldsymbol{g}(\boldsymbol{l}, \boldsymbol{x}) = \boldsymbol{0}$ between N observations $\boldsymbol{l} = (l_n)$, with covariance matrix Σ_{ll} and N unknown parameters $\boldsymbol{x} = (x_n)$, we obtain the covariance of the parameters using the implicit variance propagation law (cf. Sect. 2.7.5, p. 43),

$$\Sigma_{xx} = \mathsf{A}^{-1}\mathsf{B}\Sigma_{ll}\mathsf{B}^{\mathsf{T}}\mathsf{A}^{-\mathsf{T}} \quad \text{with} \quad \mathsf{A} = \frac{\partial \boldsymbol{g}(\boldsymbol{l},\boldsymbol{x})}{\partial \boldsymbol{x}} \quad \text{and} \quad \mathsf{B} = \frac{\partial \boldsymbol{g}(\boldsymbol{l},\boldsymbol{x})}{\partial \boldsymbol{l}} \,. \tag{12.210}$$

In our case we use this relation twice: first for determining the distances $\boldsymbol{d} = [d_1, d_2, d_3]^{\mathsf{T}}$ from the given angles $\boldsymbol{\alpha} = [\alpha; \beta; \gamma]$ and the given, fixed distances $\boldsymbol{a} = [a, b, c]^{\mathsf{T}}$ between the control points using the constraints $\boldsymbol{g}_1$, and second for the determination of the coordinates $\boldsymbol{Z} = [X_O, Y_O, Z_O]^{\mathsf{T}}$ of the projection centre from the distances and the coordinates $\boldsymbol{X}_1$, $\boldsymbol{X}_2$ and $\boldsymbol{X}_3$ of the reference points using the constraints $\boldsymbol{g}_2$:

$$\boldsymbol{g}_1(\boldsymbol{\alpha}, \boldsymbol{d}) = \begin{bmatrix} a^2 - d_2^2 - d_3^2 + 2d_2d_3\cos\alpha \\ b^2 - d_3^2 - d_1^2 + 2d_3d_1\cos\beta \\ c^2 - d_1^2 - d_2^2 + 2d_1d_2\cos\gamma \end{bmatrix}, \quad \boldsymbol{g}_2(\boldsymbol{d}, \boldsymbol{Z}) = \begin{bmatrix} d_1^2 - |\boldsymbol{Z} - \boldsymbol{X}_1|^2 \\ d_2^2 - |\boldsymbol{Z} - \boldsymbol{X}_2|^2 \\ d_3^2 - |\boldsymbol{Z} - \boldsymbol{X}_3|^2 \end{bmatrix}. \tag{12.211}$$

With the Jacobians

$$\mathsf{A}_1 = \frac{\partial \boldsymbol{g}_1}{\partial \boldsymbol{d}} = \begin{bmatrix} 0 & -2d_2 + 2d_3\cos\alpha & -2d_3 + 2d_2\cos\alpha \\ -2d_1 + 2d_3\cos\beta & & -2d_3 + 2d_1\cos\beta \\ -2d_1 + 2d_2\cos\gamma & -2d_2 + 2d_1\cos\gamma & 0 \end{bmatrix}$$

$$\mathsf{B}_1 = \frac{\partial \boldsymbol{g}_1}{\partial \boldsymbol{\alpha}} = \begin{bmatrix} -2d_2d_3\sin\alpha & 0 & 0 \\ 0 & -2d_1d_3\sin\beta & 0 \\ 0 & 0 & -2d_1d_2\sin\gamma \end{bmatrix} \tag{12.212}$$

and, similarly,

$$A_2 = \begin{bmatrix} 2(X_1 - X_O) & 2(Y_1 - Y_O) & 2(Z_1 - Z_O) \\ 2(X_2 - X_O) & 2(Y_2 - Y_O) & 2(Z_2 - Z_O) \\ 2(X_3 - X_O) & 2(Y_3 - Y_O) & 2(Z_3 - Z_O) \end{bmatrix} \quad B_2 = \begin{bmatrix} 2d_1 & 0 & 0 \\ 0 & 2d_2 & 0 \\ 0 & 0 & 2d_3 \end{bmatrix}, \tag{12.213}$$

we obtain the covariance matrix for the distances

$$\Sigma_{dd} = A_1^{-1} B_1 \Sigma_{\alpha\alpha} B_1^{\mathsf{T}} A_1^{-\mathsf{T}}, \tag{12.214}$$

and therefore the covariance matrix of the projection centre,

$$\Sigma_{ZZ} = A_2^{-1} B_2 A_1^{-1} B_1 \Sigma_{\alpha\alpha} B_1^{\mathsf{T}} A_1^{-\mathsf{T}} B_2^{\mathsf{T}} A_2^{-\mathsf{T}}. \tag{12.215}$$

Example 12.2.44: *Uncertainty of projection centre from spatial resection.* Assume the coordinates of three control points, $\mathcal{X}_1 : [-1000, 0, 0]$ m, $\mathcal{X}_2 : [500, 866, 0]$ m and $\mathcal{X}_3 : [500, -866, 0]$ m, are evenly distributed on a horizontal circle with radius 1000 m and centre, $[80, 0, 0]$ m (Fig. 12.22). The projection centre of an aerial camera with principal distance $c = 1500$ pixel is at a height of 1500 m. The image coordinates are assumed to have a precision of 0.1 pixel, which leads to standard deviations of the angles of approximately $\sigma_\alpha = \sqrt{2} \cdot 0.01/150 \approx 0.0001$ radian.

We want to discuss four positions of the projection centre with respect to the critical cylinder, and give the covariance matrix of the projection centre, the standard deviations of its coordinates, the point error $\sigma_P = \sqrt{\sigma_{X_O}^2 + \sigma_{Y_O}^2 + \sigma_{Z_O}^2}$ and the maximum loss $\sqrt{\lambda_{\max}}$ in precision with respect to an ideal situation, namely the projection centre, which is above the midpoint of the triangle; here $\lambda_{\max}$ is the maximum eigenvalue of the generalized eigenvalue problem, $|\Sigma_{ZZ} - \lambda \Sigma_{ZZ}^{\text{ref}}| = 0$ (cf. (4.263), p. 121).

1. $\mathcal{Z} : [0, 0, 1500]$ m: This is the reference situation.

$$\Sigma_{ZZ}^{(\text{ref})} = \begin{bmatrix} 0.1043 & & \\ & 0.1043 & \\ & & 0.005796 \end{bmatrix} \text{m}^2, \quad \begin{bmatrix} \sigma_{X_O} \\ \sigma_{Y_O} \\ \sigma_{Z_O} \end{bmatrix} = \begin{bmatrix} 0.32 \text{ m} \\ 0.32 \text{ m} \\ 0.076 \text{ m} \end{bmatrix}, \quad \sigma_P = 0.46 \text{ m}.$$

All coordinates can be determined quite accurately. We choose this covariance matrix to be the reference.

2. $\mathcal{Z} : [-900, 0, 1500]$ m: Here the projection centre is 100 m away from the critical cylinder, i.e.,

$$\Sigma_{ZZ} = \begin{bmatrix} 7.875 & & 6.692 \\ & 0.06064 & \\ 6.692 & & 5.699 \end{bmatrix} \text{m}^2, \quad \begin{bmatrix} \sigma_{X_O} \\ \sigma_{Y_O} \\ \sigma_{Z_O} \end{bmatrix} = \begin{bmatrix} 2.8 \text{ m} \\ 0.25 \text{ m} \\ 2.4 \text{ m} \end{bmatrix}, \quad \sigma_P = 3.7 \text{ m}, \quad \sqrt{\lambda_{\max}} = 32.$$

3. $\mathcal{Z} : [900, 0, 1500]$ m:

$$\Sigma_{ZZ} = \begin{bmatrix} 0.05858 & & -0.03301 \\ & 12.44 & \\ -0.03301 & & 0.02929 \end{bmatrix} \text{m}^2, \quad \begin{bmatrix} \sigma_{X_O} \\ \sigma_{Y_O} \\ \sigma_{Z_O} \end{bmatrix} = \begin{bmatrix} 0.24 \text{ m} \\ 3.5 \text{ m} \\ 0.17 \text{ m} \end{bmatrix}, \quad \sigma_P = 3.5 \text{ m}, \quad \sqrt{\lambda_{\max}} = 11.$$

4. $\mathcal{Z} : [1000, 900, 1500]$ m:

$$\Sigma_{ZZ} = \begin{bmatrix} 6.585 & 4.932 & -4.194 \\ 4.932 & 3.785 & -3.203 \\ -4.194 & -3.203 & 2.725 \end{bmatrix} \text{m}^2, \quad \begin{bmatrix} \sigma_{X_O} \\ \sigma_{Y_O} \\ \sigma_{Z_O} \end{bmatrix} = \begin{bmatrix} 2.6 \text{ m} \\ 1.9 \text{ m} \\ 1.7 \text{ m} \end{bmatrix}, \quad \sigma_P = 3.6 \text{ m}, \quad \sqrt{\lambda_{\max}} = 24.$$

The contour lines of $\sqrt{\lambda_{\max}}$ are given in Fig. 12.22, left. Observe that only within a region of approximately half the diameter of the circle through $(\mathcal{X}_1 \mathcal{X}_2 \mathcal{X}_3)$ is the loss in precision below a factor of 10. In Fig. 12.22, right, the situation is given for the case where point $\mathcal{X}_1 : [1000, 0, 0]$ m is chosen. The best results are achieved if the projection centre is above the midpoint of $\mathcal{X}_2$ and $\mathcal{X}_3$, although this is still approximately six times worse than the reference configuration.

Both figures confirm the critical cylinder to be a degenerate configuration. Moreover, they show how far away one needs to be from that critical situation in order to achieve a stable result. This just requires knowing the theoretical covariance matrix of the result which can be determined in *all* cases and compared to some prespecified precision. Generally, we might not have such a reference covariance matrix at hand. But it may be derived from an ideal, possibly application-dependent, distribution of the three 3D points using the knowledge of the interior orientation of the camera. This is an a posterior type of quality analysis which follows the general principles of Sect. 4.6, p. 115, and which can be realied for all other geometric estimation problems discussed in this part. We will perform such an analysis for comparing the DLT and

evaluation of final estimates

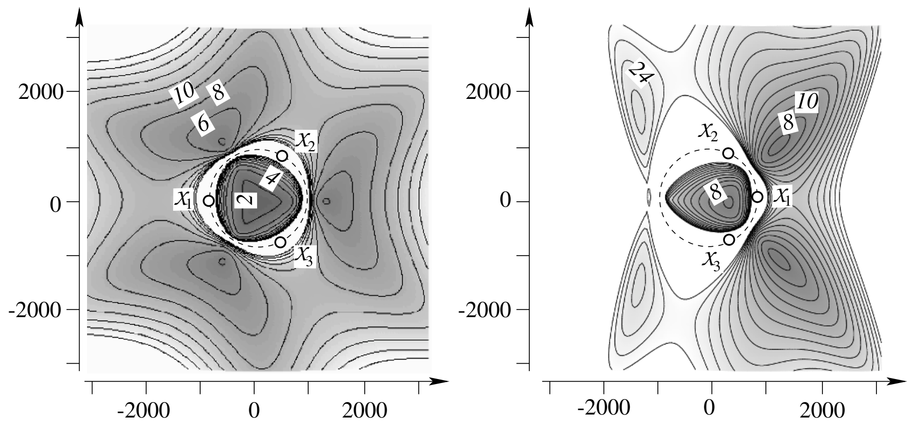

Fig. 12.22 Loss in precision for spatial resection with three given points for varying positions of the projection centre. The symmetric configuration of $\mathcal{X}_1\mathcal{X}_2\mathcal{X}_3$ with the projection centre above their midpoint serves as reference configuration. Contour lines of $\sqrt{\lambda_{\max}}$ are shown. **Left:** symmetric configuration of control points, maximum value 20. **Right:** asymmetric configuration of control points, maximum value 30; distance of isolines is 4. In the white area the values are beyond the maximum. Only at positions far off the critical cylinder can acceptable configurations be achieved: The reference configuration has the projection centre in the middle of an equilateral triangle; therefore, the minimum value of $\sqrt{\lambda_{\max}} = 1$ is at the centre of the circle. Outside the critical cylinder the best values are approximately 4, meaning that the precision is four times worse than in the reference case. In the right configuration the best value for $\sqrt{\lambda_{\max}}$ is approximately 5 and is achieved in the middle between $\mathcal{X}_2$ and $\mathcal{X}_3$

the spatial resection in an overdetermined situation and supplement it with a reliability analysis of the relative orientation of two images. $\diamond$

12.2.4.2 Direct Solution for the Spatial Resection with Four or More Points

The overdetermined spatial resection (PnP), based on more than three observed scene points or lines, though being nonlinear, has direct, i.e., noniterative, solutions. Following the review in Moreno-Noguer et al. (2007) one of the first direct algorithms is given by Quan and Lan (1999). Moreno-Noguer et al. (2007) also provide a fast algorithm, which later was extended to integrate covariance information for the image points (Ferraz et al., 2014) and to handle outliers (Ferraz et al., 2014). A solution based on Gröbner bases is given by Zheng et al. (2013).

The goal of the solution by Moreno-Noguer et al. (2007) is to derive the pose $(R, \boldsymbol{Z})$ of a calibrated camera from I pairs $(\mathcal{x}_i, \mathcal{X}_i)$ related by ${}^c\mathbf{x}'_i = d_i R(\boldsymbol{X}_i - \boldsymbol{Z})$ using a linear relationship between the given observations and some unknown parameters, similar to the DLT, cf. Sect. 12.2.2.1, p. 494. The basic idea is to represent the coordinates, $\boldsymbol{X}_i$, of the 3D points using barycentric coordinates w.r.t. four reference points $\boldsymbol{Y}_j, j = 1, ..., 4$, in the scene coordinate system. In a similar fashion we represent the camera coordinates, ${}^c\boldsymbol{X}_i = R(\boldsymbol{X}_i - \boldsymbol{Z})$, of the scene points with the same barycentric coordinates but with unknown coordinates ${}^c\boldsymbol{Y}_j$ of the reference points since the barycentric coordinates are invariant w.r.t. a spatial motion. The projection relation ${}^c\mathbf{x}'_i = d_i\, {}^c\boldsymbol{X}_i$ then can be used to determine the unknown reference coordinates ${}^c\boldsymbol{Y}_j$ in the camera coordinate system and the parameters R and $\boldsymbol{Z}$ from the now available correspondences $({}^c\boldsymbol{X}_i, \boldsymbol{X}_i)$. We describe the principle, which is the basis for the mentioned algorithm.

With the reference points $Y = [\boldsymbol{Y}_1, ..., \boldsymbol{Y}_4]$, and vectors, $\boldsymbol{\alpha}_i = [\alpha_{i1}, ..., \alpha_{i4}]^\mathsf{T}$, of the barycentric coordinates of each point we have

$$\boldsymbol{X}_i = \sum_{j=1}^{4} \alpha_{ij} \boldsymbol{Y}_j = Y \boldsymbol{\alpha}_i\,. \tag{12.216}$$

The reference coordinates are chosen such that the absolute values of the barycentric coordinates are below or around one, e.g., the centroid $\mathcal{Y}_1$ of the given points $\mathcal{X}_i, i = 1, ..., I$, and additional three points $\mathcal{Y}_j, j = 2, 3, 4$, in the direction of the three coordinate axes (XYZ) . Then the barycentric coordinates can be uniquely derived for each point using Euclideanly normalized homogeneous coordinates via $\boldsymbol{\alpha}_i = \mathsf{Y}^{-1}\mathbf{X}_i$. The same representation is used for the camera coordinates ${}^c\boldsymbol{X}_i$ of the given points, however now with unknown reference coordinates. We need two representations for these: Exercise 12.14

$$ {}^c\mathsf{Y} = [{}^c\boldsymbol{Y}_1, ..., {}^c\boldsymbol{Y}_4] \quad \text{and} \quad \boldsymbol{y} = \begin{bmatrix} \boldsymbol{y}_1 \\ \boldsymbol{y}_2 \\ \boldsymbol{y}_3 \\ \boldsymbol{y}_4 \end{bmatrix} = \operatorname{vec}\left({}^c\mathsf{Y}\right) = \begin{bmatrix} {}^c\boldsymbol{Y}_1 \\ {}^c\boldsymbol{Y}_2 \\ {}^c\boldsymbol{Y}_3 \\ {}^c\boldsymbol{Y}_4 \end{bmatrix} . \tag{12.217} $$

Thus we also have

$$ {}^c\boldsymbol{X}_i = \sum_{j=1}^{4} \alpha_{ij} {}^c\boldsymbol{Y}_j = {}^c\mathsf{Y}\boldsymbol{\alpha}_i \, . \tag{12.218} $$

Using this representation for ${}^c\boldsymbol{X}_i$, the constraint for the projection ${}^c\mathbf{x}'_i = d_i \, {}^c\boldsymbol{X}_i$ can be written as

$$ {}^c\mathbf{x}'_i \times {}^c\boldsymbol{X}_i = \mathsf{S}({}^c\mathbf{x}'_i) \, {}^c\mathsf{Y} \, \boldsymbol{\alpha}_i = \left(\boldsymbol{\alpha}_i^{\mathsf{T}} \otimes \mathsf{S}({}^c\mathbf{x}'_i)\right) \boldsymbol{y} = \underset{3\times 12}{\mathsf{M}_i^{\mathsf{T}}} \underset{12\times 1}{\boldsymbol{y}} = \underset{3\times 1}{\mathbf{0}} \, , \tag{12.219} $$

similar to (12.121) and (12.122), p. 495.

Since the barycentric coordinates, $\boldsymbol{\alpha}_i$, and the observed ray directions, ${}^c\mathbf{x}'_i$, are known, the matrices M_i are known and we can derive $\boldsymbol{y}$ by determining the eigenvector of the 12×12 matrix

$$ \mathsf{A} = \sum_{i=1}^{I} \mathsf{M}_i \mathsf{M}_i^{\mathsf{T}} \tag{12.220} $$

that is elonging to the smallest eigenvalue. This solution requires at least six points.

Let this eigenvector be $\boldsymbol{y} = \operatorname{vec}\left([{}^c\boldsymbol{Y}_1, {}^c\boldsymbol{Y}_2, {}^c\boldsymbol{Y}_3, {}^c\boldsymbol{Y}_4]\right)$. It contains the estimated camera coordinates of the four reference points, however, arbitrarily scaled, namely to 1 instead according to the scale of the reference points $\boldsymbol{Y}_j$. With (12.218) this yields arbitrarily scaled camera coordinates for all given points. Thus, from all given correspondences we solve for the sought rotation R and translation $\boldsymbol{Z}$, together with the scale γ, using the similarity transformation

$$ \gamma \, {}^c\boldsymbol{X}_i = \mathsf{R}(\boldsymbol{X}_i - \boldsymbol{Z}) \, , \quad i = 1, ..., I \tag{12.221} $$

This an be found directly using Alg. 13, p. 411.

The three constraints per observed point (12.219) are linearly dependent; thus, two independent constraints per point could be selected, e.g., replacing $\mathsf{S}({}^c\mathbf{x}'_i)$ by $\mathsf{S}^{(s)}({}^c\mathbf{x}'_i)$, cf. Sect. 7.4.1, p. 317. For perspective cameras with ${}^cw'_i \neq 0$ the first two rows can be selected as in Moreno-Noguer et al. (2007). This may slightly decrease the computational effort in (12.220).

As experiments of Moreno-Noguer et al. (2007) show, often more than one eigenvalue of A is small, however, not more than $N = 4$ (cf. Lepetit et al., 2009). Thus $\boldsymbol{y}$ is a weighted sum of the N smallest eigenvectors. This problem can be solved by integrating the estimation of the weights into the similarity transformation. While Moreno-Noguer et al. (2007) give a direct solution for the determination of N and the weighting of the N eigenvectors, Ferraz et al. (2014) propose to fix $N = 4$ and iteratively estimate the weights and thus the unknown coordinates $\boldsymbol{y}$. This way, they can handle four and more points. They also show how to handle outliers. The eigenvectors of A are then determined *robustly* which turns out to be significantly faster than RANSAC procedures. In order to arrive at a (nonrobust) maximum likelihood estimation Ferraz et al. (2014) replace the determination of the similarity transformation in (12.221) by minimizing the reprojection errors $\boldsymbol{v}^\mathsf{T}(\boldsymbol{y})\, W_{ll}\, \boldsymbol{v}(\boldsymbol{y})$ under the constraint $A\boldsymbol{y} = \mathbf{0}$.

12.2.4.3 Iterative Solution for Spatial Resection for a Central Camera

In this section we provide an iterative solution for the overconstrained spatial resection. This solution not only allows us to include very far points or points at infinity. Such scene elements may be very helpful, especially for determining the rotation of the camera. The solution also yields the covariance matrix of the resulting parameters. This can be used for evaluation of the result and for planning purposes. Last not least, the solution is the basis for the orientation of two or more images with the bundle adjustment discussed in Chap. 15, p. 643.

As we know the internal parameters of the camera, we can exploit them to derive viewing rays from observed image points or viewing planes from observed scene lines, as already done for the direct solution. Using viewing rays and planes as basic observations allows us to use the solution for the pose determination of any kind of camera with central projection.

Linearized Observation Equations for Points. Assume that we are given $I \geq 3$ 3D points $\mathcal{X}_i(\mathbf{X}_i)$, which are fixed, i.e., nonstochastic, values. They are given with homogeneous coordinates $\mathbf{X}_i = [\boldsymbol{X}_{0i}^\mathsf{T}, X_{hi}]^\mathsf{T}$, allowing for points at infinity. They are assumed to be observed, leading to camera rays $x_i'({}^c\mathbf{x}_i', \Sigma_{{}^c\mathbf{x}'_i{}^c\mathbf{x}'_i})$ represented by homogeneous coordinates and their covariance matrix. We assume all image and scene points to be conditioned and spherically normalized. For simplicity we omit the superscript ^{s} indicating the spherical normalization.

Similarly to (12.128), p. 497, the imaging model then can be written as

$$\mathbb{E}({}^c\underline{\mathbf{x}}_i') = \mathsf{N}(\tilde{R}[I_3 \mid -\tilde{\boldsymbol{Z}}]\, \mathbf{X}_i)\,, \quad \mathbb{D}({}^c\underline{\mathbf{x}}_i') = \Sigma_{{}^c\mathbf{x}_i'{}^c\mathbf{x}_i'}\,, \quad i = 1, ..., I\,, \tag{12.222}$$

where we used the normalization operator $\mathsf{N}(\mathbf{x}) = \mathbf{x}/|\mathbf{x}|$ for fixing the scale.

This is a nonlinear Gauss–Markov model. We have two observations per image point, so that we need to reduce the three constraints to two. If we have approximate values ${}^c\widehat{\mathbf{x}}_i'^a$ for the camera rays we can project the constraints onto the tangent space of the spherically normalized camera rays and obtain the nonlinear constraints (cf. Sect. 10.2.2.1, p. 369)

$$\Delta{}^c\boldsymbol{x}_{ri}' + \widehat{\boldsymbol{v}}_i = J_r^\mathsf{T}({}^c\widehat{\mathbf{x}}_i'^a)\, \mathsf{N}(\widehat{R}[I_3 \mid -\widehat{\boldsymbol{Z}}]\, \mathbf{X}_i)\,, \quad \mathbb{D}({}^c\underline{\mathbf{x}}_i') = \Sigma_{{}^c\mathbf{x}_i'{}^c\mathbf{x}_i'} \quad i = 1, ..., I\,, \tag{12.223}$$

with the reduced observations

$$\Delta{}^c\boldsymbol{x}_{ri}' = J_r^\mathsf{T}({}^c\widehat{\mathbf{x}}_i'^a){}^c\mathbf{x}_i' \quad \text{with} \quad J_r(\mathbf{x}) = \mathsf{null}(\mathbf{x}^\mathsf{T})\,. \tag{12.224}$$

Using the multiplicative update of rotations $R = R(\Delta\boldsymbol{r})R^a$ and writing the argument of the normalized vector as $\widehat{R}[I_3 \mid -\widehat{\boldsymbol{Z}}]\, \mathbf{X}_i = \widehat{R}(\boldsymbol{X}_{0i} - X_{hi}\widehat{\boldsymbol{Z}})$ yields the observation equations for the linear substitute Gauss–Markov model, now making the spherical normalized vector ${}^c\mathbf{x}_i'^s$ explicit,

$$\begin{aligned}
\Delta{}^c\boldsymbol{x}_{ri}' + \widehat{\boldsymbol{v}}_i &= \frac{\partial{}^c\boldsymbol{x}_{ri}'}{\partial{}^c\mathbf{x}_i'^s}\frac{\partial{}^c\mathbf{x}_i'^s}{\partial{}^c\mathbf{x}_i'}\left(\frac{\partial{}^c\mathbf{x}_i'^s}{\partial\boldsymbol{r}}\widehat{\Delta\boldsymbol{r}} + \frac{\partial{}^c\mathbf{x}_i'^s}{\partial\boldsymbol{Z}}\widehat{\Delta\boldsymbol{Z}}\right) &(12.225)\\
&= \underbrace{J_r^\mathsf{T}({}^c\widehat{\mathbf{x}}_i'^a)\, J_s({}^c\widehat{\mathbf{x}}_i'^a)\left[-X_{hi}\widehat{R}^a \mid -S\left(\widehat{R}^a(\boldsymbol{X}_{0i} - X_{hi}\widehat{\boldsymbol{Z}}^a)\right)\right]}_{A_i^\mathsf{T}}\begin{bmatrix}\widehat{\Delta\boldsymbol{Z}}\\ \widehat{\Delta\boldsymbol{r}}\end{bmatrix}\,, &(12.226)
\end{aligned}$$

and $J_s(\mathbf{x}) = (I_3 - \mathbf{x}\mathbf{x}^\mathsf{T}/|\mathbf{x}|^2))/|\mathbf{x}|$, cf. (10.18), p. 368. This linearized observation equation with the 2×6 Jacobian A_i^T per image point can be used within an iterative estimation scheme.

Exercise 12.13

critical configuration

There exists a critical configuration: If all points, including the projection centre, lie on a horopter curve (Buchanan, 1988; Wrobel, 2001; Hartley and Zisserman, 2000; Faugeras and Luong, 2001), the solution is ambiguous.

observed scene lines

For observed scene lines $\mathcal{L}_j(\mathbf{L}_j)$ we can use the following model:

$$\mathbb{E}({}^c\underline{\mathbf{l}}_j'^s) = \mathsf{N}\left(\widehat{R}\,[S^\mathsf{T}(\widehat{\boldsymbol{Z}}) \mid I_3])\,\mathbf{L}_j\right), \quad \mathbb{D}({}^c\underline{\mathbf{l}}_j'^s) = \Sigma_{\mathbf{l}_j'^s\mathbf{l}_j'^s}, \quad j = 1, \ldots, J, \tag{12.227}$$

cf. (12.76), p. 481 and Luxen and Förstner (2001); Ansar and Daniilidis (2002); Mirzaei and Roumeliotis (2011).

12.2.5 Theoretical Precision of Pose Estimation

We give the theoretical precision of the orientation parameters for a standard case. We parametrize the configuration, and, as for the direct solution of the spatial resection, analyse the theoretical precision algebraically.

We assume a block with eight corners is observed with a nadir view (cf. Fig. 12.23). We parametrize the block in relation to the height Z_O of the projection centre above the

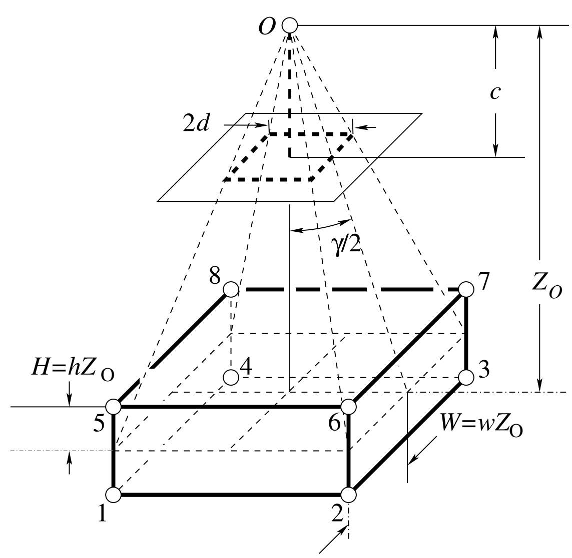

Fig. 12.23 Eight point configuration for the orientation of a camera

centre of the block and the two factors h and w defining the height and the width of the block. Referring to the centre of the block, the height of the projection centre is Z_O, the principal distance of the camera is c, and the width and depth of the block is $2W = 2wZ_O$, related to the angle γ under which the block is visible by $W = Z_O \tan(\gamma/2)$. The height of the block is $2H = 2hZ_O$. Observe that $h < 1$, as otherwise the projection centre is inside the block. The image coordinates are assumed to be measured with a standard deviation of $\sigma_0 = \sigma_{x'} = \sigma_{y'}$.

12.2.5.1 Theoretical Precision Using an Uncalibrated Camera

We start with the pose determination using an uncalibrated camera. From the observation equations (12.120), p. 495 we can algebraically derive the normal equation matrix and its inverse using an algebra package such as MAPLE. We obtain the theoretical standard deviations for the exterior orientation, for scale difference m, and for shear s between the sensor coordinate axes,

$$
\begin{aligned}
\sigma_{X_O} = \sigma_{Y_O} &= \frac{\sqrt{2}}{4}\frac{|1-h^2|}{h}\sqrt{\frac{(1+h^2)^2+8h^2}{(1+h^2)^2+4h^2}}\frac{Z_O}{c}\sigma_0 \\
\sigma_{Z_O} &= \frac{1}{4}\frac{|1-h^2|}{h}\frac{\sqrt{1+h^2}}{h}\frac{Z_O}{c}\sigma_0 \\
\sigma_\omega = \sigma_\phi &= \frac{\sqrt{2}}{4}\frac{(1-h^2)^2}{\sqrt{(1+h^2)^2+4h^2}}\frac{1}{c}\sigma_0 \\
\sigma_\kappa = \frac{1}{2}\sigma_m = \frac{1}{2}\sigma_s &= \frac{\sqrt{2}}{4}\frac{1}{w}\frac{|1-h^2|}{\sqrt{1+h^2}}\frac{1}{c}\sigma_0 \,.
\end{aligned}
$$

The expressions for the other parameters of the interior orientation are lengthy and therefore not given. The height factor $h > 0$ of the block needs to be nonzero for the projection centre to be determinable, confirming the critical configuration when all points are coplanar. The precision of the coordinates of the projection centre is given in Fig. 12.2.5.1. Obviously, the precision of the Z_O coordinate is always worse than the precision of the X_O and Y_O coordinates.

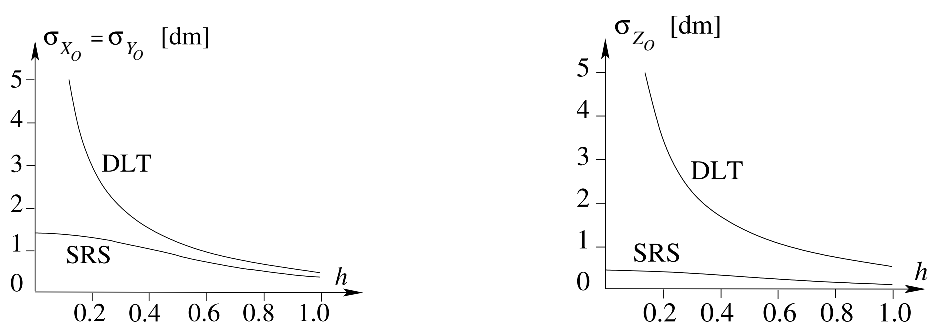

Fig. 12.24 Theoretical precision of the projection centre from the orientation of a camera given the observations of the eight corners of a block. Flying height over ground 1500 m, principal distance $c =$ 1500 pixel, $\sigma_0 = 0.1$ pixel, half the width of the image of the block $d = 9.2$ cm. **Left:** precision in XY. **Right:** precision in Z. DLT: direct estimation of the projection matrix with DLT, i.e., 11 parameters for a straight line-preserving camera, SRS: spatial resection, i.e., six parameters for a calibrated camera. The factor h is the ratio H/Z_O of the height of the block in units to the distance Z_O

The large standard deviations for the projection centre result from large correlations with the rotation angles and the principal distance. We have

$$
\rho_{X_O\phi} = -\rho_{Y_O\omega} = \frac{-2h}{\sqrt{(1+h^2)^2+8h^4}}\,, \quad \rho_{Z_Oc} = \frac{1+3h^2}{\sqrt{1+8h^2+5h^4+2h^6}}\,. \qquad (12.228)
$$

The maximal correlation $|\rho_{X_O\phi}|$ between X_O and ϕ is 0.5, which is acceptable. For heights H of the block larger than half of the flying height Z_O, the correlation between the position X_O of the projection centre and the principal point x'_H is less than 0.9. However, only for values $h > 1.5$, which is rarely used, is the correlation $|\rho_{Z_Oc}|$ of the distance Z_O between object and camera and the principal distance c better than 0.95.

12.2.5.2 Theoretical Precision Using a Calibrated Camera

We now give the theoretical precision for the exterior orientation of a vertical view using a calibrated camera. Unfortunately, the expressions for the standard deviations are lengthy. For simplification, we therefore assume all points to lie in one plane; thus, the height of the block is assumed to be 0. This corresponds to measuring *double points in the four corners of a square*. Its size in the image is $2d \times 2d$ (see Fig. 12.23, p. 521). We obtain the following expressions for the standard deviations:

$$\sigma_{X_O} = \sigma_{Y_O} = \frac{\sqrt{2}}{4}\frac{Z_O}{c}\sqrt{1+\frac{1}{\sin^4\frac{\gamma}{2}}} \quad \sigma_0 \tag{12.229}$$

$$\sigma_{Z_O} = \frac{Z_O}{4d} \quad \sigma_0 \tag{12.230}$$

$$\sigma_\omega = \sigma_\phi = \frac{\sqrt{2}}{4}\frac{c}{d^2} \quad \sigma_0 \tag{12.231}$$

$$\sigma_\kappa = \frac{1}{4d} \quad \sigma_0 \,. \tag{12.232}$$

We now just have correlations between the lateral position (X_O, Y_O) and the angles ω and ϕ, namely

$$\rho_{X_O\phi} = -\rho_{Y_O\omega} = \frac{1}{\sqrt{1+\sin^4\frac{\gamma}{2}}} \,. \tag{12.233}$$

For viewing angles of $\gamma = 90°$ the correlations approach 0.9, while for viewing angles below 45^o the correlations are nearly 0.99. The standard deviation of the lateral position of the camera determined with a spatial resection is always larger than the standard deviation of the distance between object and camera by a factor larger than 3.1.

Compared to the precision obtainable with the DLT, the superiority of the spatial resection is obvious: In order to obtain the same height accuracy σ_{Z_O} as with the spatial resection solution, we need to exploit the full depth, namely using a cube with side length Z_O. In order to obtain the same lateral precision $\sigma_{X_O} = \sigma_{Y_O}$ we need to have a spatial object with a height of at least $H > 0.6\, Z_O$ when using the DLT. On the other hand, with the DLT, already small height differences H lead to standard deviations which are only approximately larger, by a factor Z_0/H, than those of the spatial resection. This might be sufficient in certain applications.

precision of DLT is worse than that of spatial resection

precision of DLT increases with relative height variations

12.3 Inverse Perspective and 3D Information from a Single Image

Inverse perspective covers all aspects of recovering partial information of the scene or of the camera from a single image. Methods of inverse perspective have been extensively studied by Criminisi (2001), including all types of 3D measurements. We provide methods of inverse perspective based on image points, lines or conics, illustrate the use of the cross ratio, and finally present a method for recovering the pose of a 3D circle from its image. For the reconstruction of 3D shapes from line drawings, cf. Sugihara (1986); Cooper (1993); Varley and Martin (2002).